

Which Features Are Most Predictive of Hiring Outcomes: Experience, Skills, or Education?

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Dataset:

StackOverflow Developer Survey 2024

GitHub Repository:

[<https://github.com/mchikovaniieu2024-create/ML-fundamentals-2026-Final-Project-Group5.git>]

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1. Problem Statement and Objective

This project showcases employability prediction as a supervised binary classification task using developer survey data from StackOverflow.

The research question we are basing this project on is: **Which Features Are Most Predictive of Hiring Outcomes: Experience, Skills, or Education?** We formulated it on the fact that employability-related datasets are often noisy, high-dimensional, and contain complex relationships between variables. More specifically, factors such as education, experience, skills, and specializations can interact in many, potentially non-linear ways, making prediction difficult and unclear. As such, the objective of the project is to use these factors to predict the target variable, **Employed**, where a value of 1 represents employed respondents and 0 represents non-employed respondents, serving as the key thread.

The motivation for the project is to better understand which factors contribute most to employability. It also supports workforce research by helping better understand patterns associated with employment outcomes. It also aims to be insightful for students, job seekers, and educators alike to understand which factors are most strongly associated with employment status.

2. Dataset Summary and Preprocessing

The dataset is the Stack Overflow Developer Survey 2024, containing 73,462 rows and 15 columns before cleaning, reduced to 12 columns after preprocessing. The target variable, **Employed**, is reasonably balanced: 39,392 employed respondents (53.6%) versus 34,070 not employed (46.4%). This balance meant accuracy remained a useful metric alongside F1 and ROC-AUC.

Dropped columns

Three columns were dropped. **Unnamed** was a leftover CSV index with no predictive value. **PreviousSalary** was removed because knowing a previous salary implies the person was already employed, which encodes the outcome we are trying to predict. **Employment** was dropped for the same reason: it is a direct categorical encoding of employment type, making it a trivial proxy for the target.

Missing values

The only column with missing values was **HaveWorkedWith**, which had 63 nulls. These were filled with empty strings, treating a missing entry as a respondent who listed no technologies.

Leakage prevention

The dataset was split into training, validation, and test, using a stratified 70/15/15 split. Stratification was applied on `Employed` to preserve the class ratio across all three sets. Validation results were used for model comparison during development; the test set was held out and used only for final reporting.

The train/validation/test split was performed on the raw, cleaned data, before any encoding or scaling. All transformers were fitted exclusively on the training set and applied to validation and test using `transform()` only.

3. Pipeline and Feature Engineering

3.1 Baseline Features

The baseline preprocessing pipeline was implemented in the `data_utils.py` file. Baseline was created as a simple benchmark only taking into account basic experience and education variables, excluding the skills representations, as further on we would evaluate whether advanced skill variables improve the performance.

The variables included were `EdLevel`, `Accessibility`, `MentalHealth`, `MainBranch`, `Gender`, `Age`, `YearsCode`, and `YearsCodePro`. `EdLevel` was ordinal encoded as it has a hierarchical order, the rest of the categorical variables were one-hot encoded as they do not have a numerical order. `YearsCode` and `YearsCodePro` were normalised using `StandardScaler` so that larger values do not dominate the model. All the skill-based features, such as `HaveWorkedWith`, were excluded for comparison purposes later in the analysis.

3.2 Education Features

The education preprocessing pipeline can be seen in the `data_utils.py` file. The education feature group focuses primarily on education variables, but it also takes into account demographic and background-related information as they provide a more complete respondent profile, improving the analysis.

The variables used were `EdLevel`, `Accessibility`, `MentalHealth`, `MainBranch`, `Gender`, and `Age`. As mentioned before, `EdLevel` was ordinal encoded, and the rest of the categorical variables were one-hot encoded. This feature group assessed the predictive power of education-related variables independently of skill representations and experience variables.

3.3 Experience Features

For the Experience model, for which the code is available in `data_utils.py`, we are using `YearsCode` and `YearsCodePro` feature groups to represent the coding and professional programming experience of the respondents. We used the `StandardScaler` to standardize these numerical variables to ensure that we could compare the features at the same magnitudes during model training.

However, these features do not capture the broad real-world professional experiences, they mainly portray the coding duration. Factors such as seniority, specialization, project quality, internships, and transferable skills are not represented, which became very important later in the project when interpreting the weak performance of the experience-only model.

3.4 Skills Features

Skills feature engineered on the `HaveWorkedWith` column. Skills were parsed and normalized beforehand. Code available in `process_skills.py` and `feature_engineering.py`.

3.4.1 Counts

The `SkillCountTransformer` function returns `total_skills`, the number of distinct skills. It treats all technical skills as equivalent.

3.4.2 Binary

The `BinarySkillTransformer` function used the `SkillVocabulary` to learn a binary vocabulary from training data. This creates a sparse binary skill matrix where each skill becomes a separate column. We set this parameter(`min_freq = 2`) so that skills that only appeared once did not have their own column.

3.4.3 Grouped

`GroupedSkillTransformer` function maps technologies into :

- frontend
- backend
- databases
- cloud
- DevOps
- programming languages
- data science
- Mobile

Higher-level groups were selected from a frequency distribution of technology skills. The output is a count and a binary “has group” column for each group.

3.4.4 Embeddings

The `EmbeddingTransformer` function receives a parsed skill list from the function `build_skill_text`. It uses the `SentenceTransformer("all-MiniLM-L6-v2")` model from the sentence-transformer library. This encodes these strings into dense semantic numerical vectors which captures similarity and meaning of skills.

4. Models Used and Where

Five models were tested. Logistic regression and random forest were applied in education-only, experience-only, and skills-only models. Gradient boosting was applied to the combined model..

<i>Model</i>	<i>Purpose</i>
Zero-rule baseline	Provided the baseline of the performance range by predicting the majority class for every example, achieving a ROC-AUC score of 0.5.
Baseline Logistic Regression	Trained using education and experience features to provide a simple linear comparison to the other models.
Logistic Regression	Applied across each feature-group combination, providing an interpretable linear representation of the data.
Random Forest	Evaluated across education, experience, and skills feature groups because it can model more complex relationships between features.
Gradient Boosting	Reserved for the combined model experiment, where the larger feature space justified training a more complex nonlinear model.

Table 1. Models evaluated throughout the experiments and their roles within the pipeline.

All the models were evaluated based on Accuracy, Precision, Recall, F1 score, and ROC-AUC. ROC-AUC was important since it evaluates how well a classifier separates the two classes from each other, regardless of the 0.5 classification threshold. It can be used to detect poorly discriminating models producing biased results in terms of precision, recall, or both, at least depending on some arbitrary choice of the threshold.

One honest constraint worth stating: the project used a fixed split rather than cross-validation. Given the dataset size and the cost of running multiple embedding-based pipelines, full cross-validation was not practical here. Hyperparameters were set manually at reasonable values rather than through grid search. This means the reported metrics reflect a single split, and variance across splits is unknown, a limitation that matters most for the binary skills model, where generalization beyond this dataset is already uncertain.

5. Key Results

Experiment	Model	Val Acc	Val Prec	Val Rec	Val F1	Val AUC	Test Acc	Test Prec	Test Rec	Test F1	Test AUC
Skills	binary_lr	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Skills	binary_rf	0.993	0.991	0.997	0.994	1.000	0.995	0.993	0.997	0.995	1.000
Combined	combined_lr	0.953	0.954	0.958	0.956	0.991	0.952	0.953	0.958	0.955	0.992
Skills	embeddings_lr	0.945	0.945	0.953	0.949	0.989	0.947	0.947	0.955	0.951	0.990
Combined	combined_gb	0.904	0.904	0.918	0.911	0.972	0.909	0.910	0.921	0.916	0.974
Skills	embeddings_rf	0.883	0.892	0.889	0.891	0.957	0.886	0.895	0.892	0.893	0.960
Skills	grouped_lr	0.834	0.840	0.854	0.847	0.918	0.834	0.842	0.851	0.846	0.921
Skills	grouped_rf	0.812	0.821	0.832	0.826	0.897	0.814	0.823	0.832	0.828	0.897
Skills	counts_rf	0.783	0.783	0.823	0.803	0.868	0.786	0.787	0.824	0.805	0.872
Skills	counts_lr	0.777	0.811	0.761	0.785	0.868	0.780	0.817	0.760	0.788	0.872
Experience	experience_lr	0.536	0.536	0.995	0.697	0.505	0.537	0.537	0.997	0.698	0.491
Experience	experience_rf	0.539	0.547	0.813	0.654	0.523	0.537	0.546	0.809	0.652	0.518
Education	education_lr	0.569	0.561	0.91	0.694	0.57	0.565	0.559	0.898	0.689	0.569
Education	education_rf	0.568	0.571	0.788	0.662	0.574	0.563	0.568	0.774	0.655	0.567

Table 2. Comparison metrics table for all models.

Validation and test results are closely aligned across models, suggesting a consistent held-out performance and no validation-only effect.

5.1 Baseline Model

Zero-rule baseline achieved an F1 score of 0.6981 and ROC-AUC value of 0.5, indicating that the model is guessing the classes randomly. The Logistic Regression model results were slightly better than the zero-rule baseline =F1 score of 0.6877 and an ROC-AUC score of 0.5652. The ROC-AUC score of the Logistic Regression was higher, suggesting that the model was able to learn some of the patterns in the education and experience variables.

The overall results were weak; the baseline model is unable to separate the classes effectively and should be used as a reference benchmark for comparison when more advanced variables such as skills are introduced.

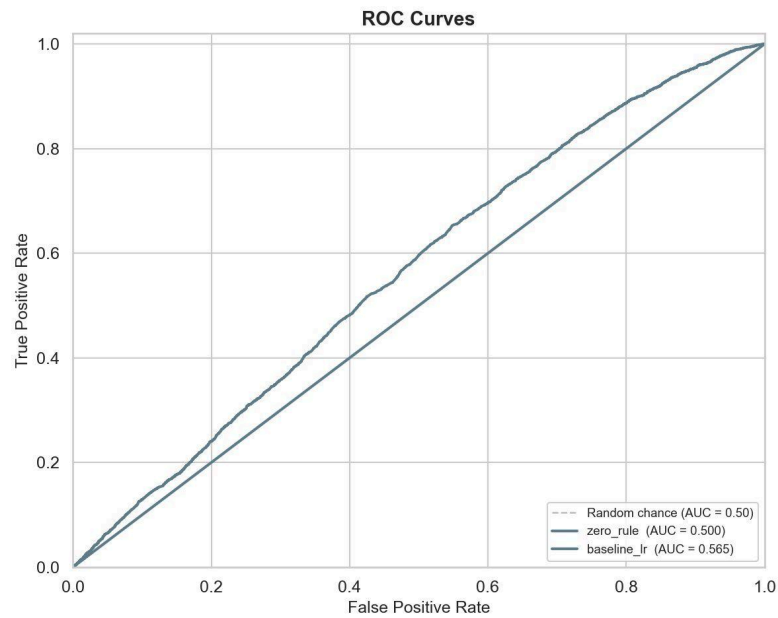


Figure 1. ROC curve of the zero-rule baseline and logistic regression. As can be seen from the graph, the logistic regression is better at class separation but overall, the result is weak.

5.2 Education Model

The education-only model performed very similarly to the reference benchmark, indicating weak standalone predictive power. The Logistic Regression had an F1 score of 0.655 and ROC-AUC value of 0.567, Random Forest had an F1 score of 0.655 and ROC-AUC value of 0.567.

As their ROC-AUC scores were not significantly better than random guessing, both models performed poorly on class separation. The results show that education-only variables contained limited predictive signals.

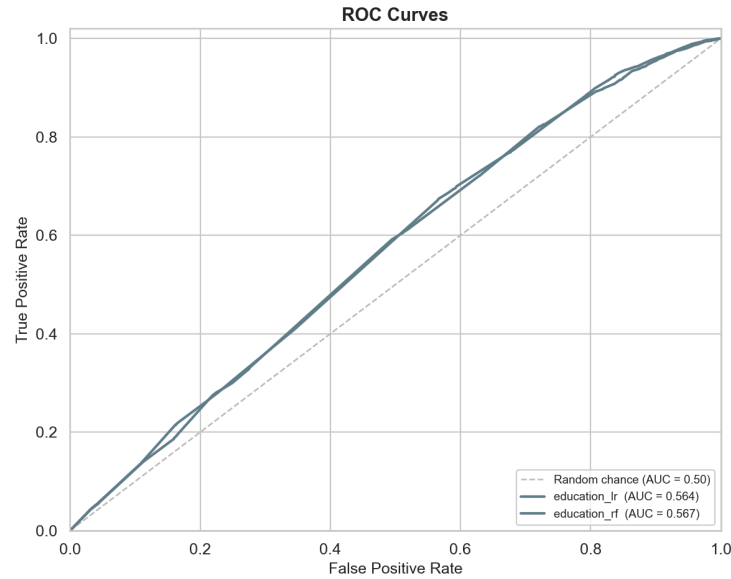


Figure 2. ROC curve of education logistic regression and random forest. As can be seen from the graph the curves are very close to the threshold implying weak class separation.

5.3 Experience Model

The experience-only model performed poorly. Logistic Regression was used as the main example because it achieved the highest recall of 0.9968, while Random Forest had 0.8008. Logistic Regression produced values below 0.5 for the ROC-AUC curve. This showed weak class separation and limited predictive power. This suggests that `YearsCode` and `YearsCodePro` were too narrow to represent the real-world professional experience.

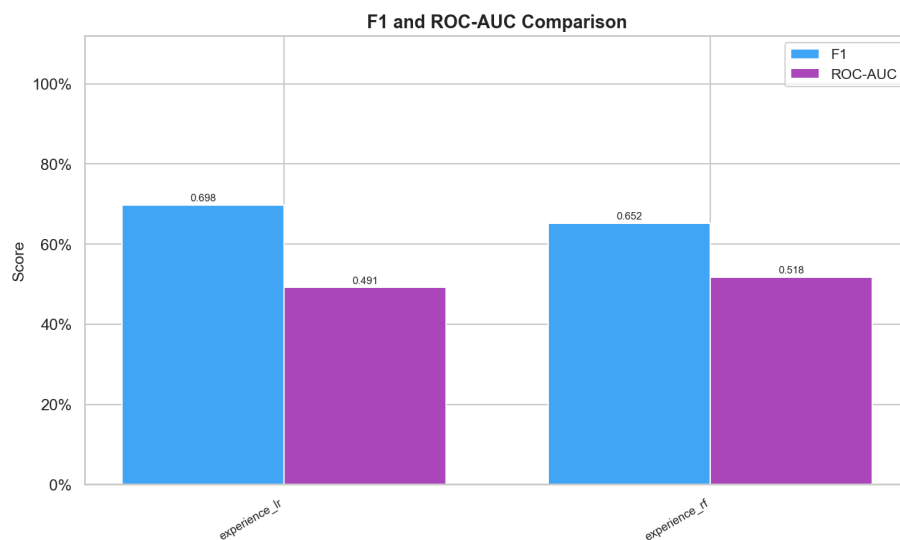


Figure 3. *F1 and ROC-AUC comparison for the experience-only models.* Both models showed weak predictive performance despite having a relatively high recall.

5.4 Skills Model

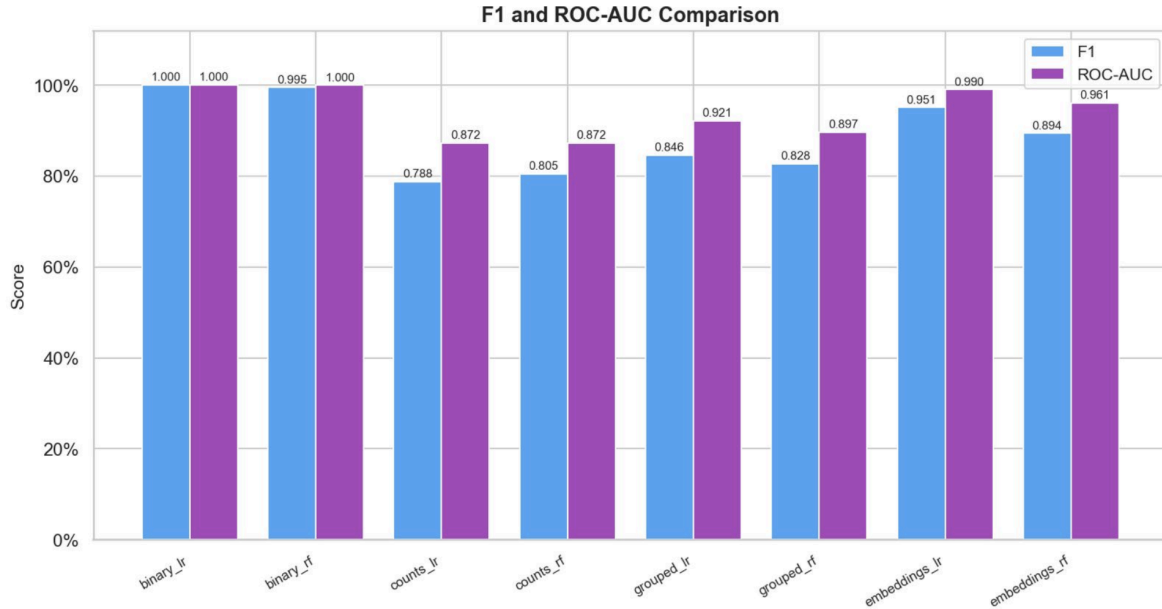


Figure 4. *F1 and ROC-AUC comparison for the skills models.* All models showed high predictive power.

Logistic regression models gained higher scores than random forest. Our feature representation made the classes linearly separable. Complexity did not add to the model performance.

For logistic models:

The binary representation = ROC-AUC and F1 score of 1.00 , a suspiciously perfect score treated with caution. The count's representation = ROC-AUC score of 0.872, still meaningful, but the most limited. Grouped representations = ROC-AUC score of 0.921, improving performance due to a higher technical structure. Sentence embeddings = ROC-AUC score of 0.990 and F1 score of 0.951, the semantic meanings allowed the model to capture most of the predictive power.

The skills dominated, and even the weakest skill representation (counts) performed better than experience-only or education-only. The binary model raises concerns in data leakage and generalisability.

5.5 Combined Model

The final combined model integrated semantic embeddings, education, and experience features. Logistic Regression outperformed Gradient Boosting. The combined embedding-based Logistic Regression model achieved approximately ROC-AUC ≈ 0.992 which is higher than the Random Forest ROC-AUC ≈ 0.974 .

As previously mentioned, the embedding-based representation produced a feature space that was already highly separable, hence, leaving less nonlinear structure for Gradient Boosting to exploit. The model retained the strong predictive signal of the embeddings while reducing the over-specific behaviour seen in the binary skills model.

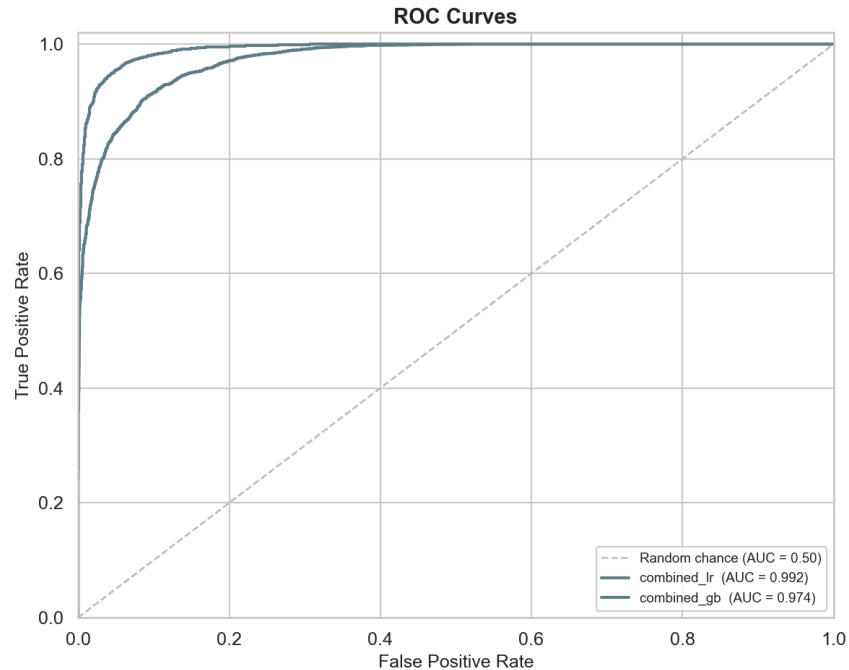


Figure 5. ROC Curve Comparison for Combined Feature Models. Logistic Regression achieved stronger class separation than Gradient Boosting when combining embeddings, education, and experience features.

6. Binary Skills Diagnostic

Near-perfect metrics in the binary logistic and random forest model raised serious concerns in:

- Data leakage
- Data memorisation / overfitting

We underwent data leakage checks, and a shuffled labelled diagnostic.

Shuffled-label diagnostic

Metric	Value
Validation Accuracy	0.5129
ROC-AUC	0.4889

Table 3. Shuffled-label Diagnostic.

This shows random levels were achieved, meaning no obvious pipeline leakage. Our understanding of the near e34reñl-perfect scores is that binary skills created exact yes/no columns per skill, making applicants perfectly separable. This is why embeddings were more appropriate for the combined model, due to overfitting concerns with external data using the binary model. The binary skills would also dominate the experience and education features, whose combinatory predictive power we wanted to explore.

7. Limitations, Failures, and Reflection

7.1 Limitations

A lack of external validation means we had no different independent test set to test our claims, which is further extended on to the issue of no cross-validation, in which we could see if the findings we had were generalizable.

We used embeddings which were a trade-off for interpretability, as they could capture complex relationships between different aspects of the data, however as a result, the relationships captured by the embeddings were difficult to interpret. Moreover, there was no systematic hyperparameter tuning, hyperparameters were selected manually rather than through systematic optimization methods such as grid search, for example. Also, the experience features were too narrow, which means they measured the quantity of experience (i.e., years) rather than its quality or relevance.

Next, the binary skills for certain things like Python, Pytorch, Docker, etc., might have been very dataset specific and difficult to generalize to other datasets and particularly outside IT-related employment contexts. Lastly, an extension to the work could be to investigate whether the inclusion of demographic features introduces bias into the education model and how these variables affect predictive hiring outcomes across different groups.

7.2 Failures / Negative Results

Considering failures and poor results, the education-only models showed relatively weak predictive performance, suggesting that education alone may not be sufficient for employability prediction. Similarly, the mentioned experience-only models produced unstable results, possibly due to the narrow representation of experience-related features being only weighted by years/duration of experience, and not quality and/or relevance.

Lastly, the binary skill models may have become overly dataset-specific, limiting their generalizability to other contexts. These were not failures per se, but more specifically, some drawbacks of our model that made the end results a bit less universal and applicable to other domains, and were to be taken with caution.

7.3 Ethical considerations

In regard to ethics, this project raised valid ethical considerations related to structural inequality, demographic sensitivity, bias risk, fairness and universality in predictive hiring models. Since demographic and professional features may reflect existing social biases, the model could unintentionally reinforce these hiring patterns, undermining the purpose of developing an automation like this. Therefore, the model should not be used directly for hiring decisions without ensuring fairness through external validation, to ensure reliable and proper use of the solution.

8. Group Collaboration and Task Division

The project was divided by module ownership to enable parallel development through GitHub. Each member committed their updates to the main branch.

- Massimo: data preprocessing, cleaning pipelines, and project configuration.
- Avisia: implementing the baseline model and education-only models.
- Jade: feature engineering of skills and implementing all skill models.
- Marie: built the experience-only and combined model.
- Maya: evaluation metrics and plots.
- Faris: developed the main script.

9. Conclusion

To conclude, the strongest predictive performance was achieved by semantic skill embeddings. The combined embedding-based Logistic Regression demonstrated the strongest balance between robustness, predictive performance and generalization. One of the key insights from our analysis was that feature representation was more important than the complexity of the model, which is why after transforming the feature space effectively using embeddings, Logistic Regression outperformed more complex methods. Even though education and experience had weak predictive power alone, they still added some predictive information to the combined model.

Furthermore, through the analysis we learned that evaluation of high-performance metrics is very important and we shouldn't trust perfect scores right away without further investigation, which is why in our project we explored potential leakage, overfitting and generalisability risks to better understand the models and what they are learning.

For our next steps we would include more in-depth experience representations, integrate temporal validation to better evaluate whether historical patterns are still reliable after distribution shifts and test the models on external datasets.